Humanoid Robots in Austrian Intralogistics: A Cost Decision Memo

Executive Summary

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Recommendation

Act now: deploy Lean Human + AMR Team (1 human + 3 AMRs, ρ≤0.85). At €453K in five-year present-value cost, it is the least-cost modeled capacity-feasible option. **Lean Human Team** (3 people, €647K) remains the human-only comparator. **For humanoid-inclusive alternatives:** Future 2028 Humanoid Team is lowest at €2.533M, but remains costlier than the human baseline. Do not approve humanoid procurement solely on wage-arbitrage assumptions; local vacancies and warehouse telemetry are the decision-critical validation inputs.

Decision at a glance

Scenario	5-year NPV	Total-cost reduction vs baseline (NPV)	Decision
Lean Human + AMR Team	**−€453K**	**73.8%**	Implement now
Lean Human Team	−€647K	62.5%	Human-only comparator
Legacy Overstaffed Team	−€1.726M	0.0%	Legacy reference
Future 2028 Humanoid Team	**−€2.533M**	**−46.7%**	Lowest-cost humanoid case
All-Humanoid Team	−€2.878M	−66.7%	Not preferred
Human + Humanoid + AMR Team	−€3.841M	−122.5%	Integration burden

All values come from `tco_scenarios.csv`. They are cost NPVs, so a less-negative number represents lower cost.

Scope and evidence

The project combines 2,359 publicly available UnifoLM demonstration episodes, a SimPy AutoStore model, and a five-year TCO model using Austrian labor inputs. The data are household demonstrations rather than warehouse tasks; the model therefore applies a 0.50–0.90 WBT-to-production transfer factor.

Method

1. Capability extraction summarizes cycle-time evidence.
2. Discrete-event simulation checks operational feasibility.

3. The financial model calculates capex, recurring operating costs, residual value, and the five-year cost NPV.
4. OAT, paired Monte Carlo (common random numbers), and two-way frontiers test sensitivity without changing scenario identity.

Key limitations

Availability derating, wage growth, system integration (€200K for robot scenarios), and residual value are modeled. They are assumptions, not measurements. The published frontiers do not identify a Future 2028 Humanoid Team / Lean Human + AMR Team cost tie within the €60K–€180K humanoid-capex grid. A local pilot should replace the transfer proxy before a capital commitment.